

Student Performance Prediction using ML

A Random Forest Classifier Approach for Early Academic Intervention

 Machine Learning Project Analysis

Executive Summary

This project focuses on predicting student academic success based on socio-economic and demographic factors. By leveraging a dataset of 1,000 students and implementing a **Random Forest Classifier**, we achieved an exceptional **98% accuracy** in classifying students into performance categories.

The Need for Predictive Analytics

Early Warning Systems

Identifying at-risk students before they fail is a critical challenge in modern education.

Machine Learning allows institutions to analyze complex patterns in historical data to provide proactive support and personalized intervention strategies.



Project Objectives

-  **Data Preprocessing:** Clean and transform categorical student data into numerical formats.
-  **Feature Engineering:** Construct a unified performance metric based on cross-subject average scores.
-  **Model Training:** Implement Random Forest algorithm to manage high-variance educational data.
-  **Evaluation:** Validate model robustness using accuracy, precision, and confusion matrices.

Development Environment



Software Stack

- Python 3.x Environment
- Google Colab / Jupyter
- Pandas & Scikit-Learn
- Matplotlib & Seaborn



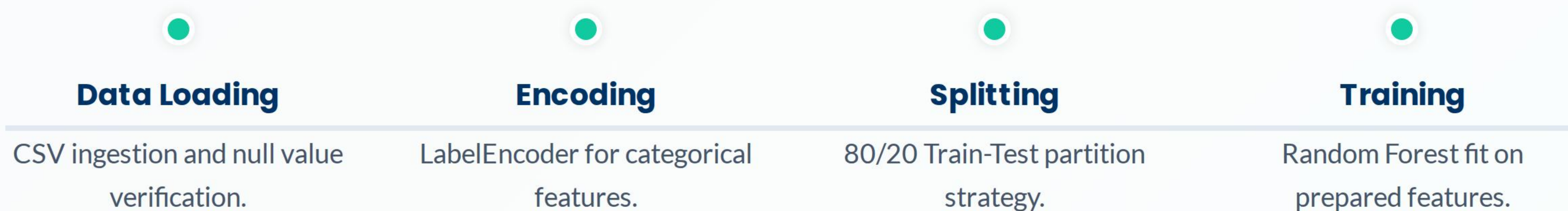
Hardware Specs

- Minimum 8GB RAM
- Multicore Processor (i5+)
- Integrated GPU support
- Cloud-based execution

Dataset Attributes

Feature Category	Attributes Included	Data Type
Demographics	Gender, Race/Ethnicity	Categorical
Socio-Economic	Parental Education, Lunch Type	Categorical
Preparation	Test Prep Course Completion	Binary
Performance	Math, Reading, Writing Scores	Integer (0-100)

The Analytical Pipeline



$$\text{Average Score} = \frac{\text{Math} + \text{Reading} + \text{Writing}}{3}$$

| Target Variable Distribution

540

Class 0 (Fail/Avg)

460

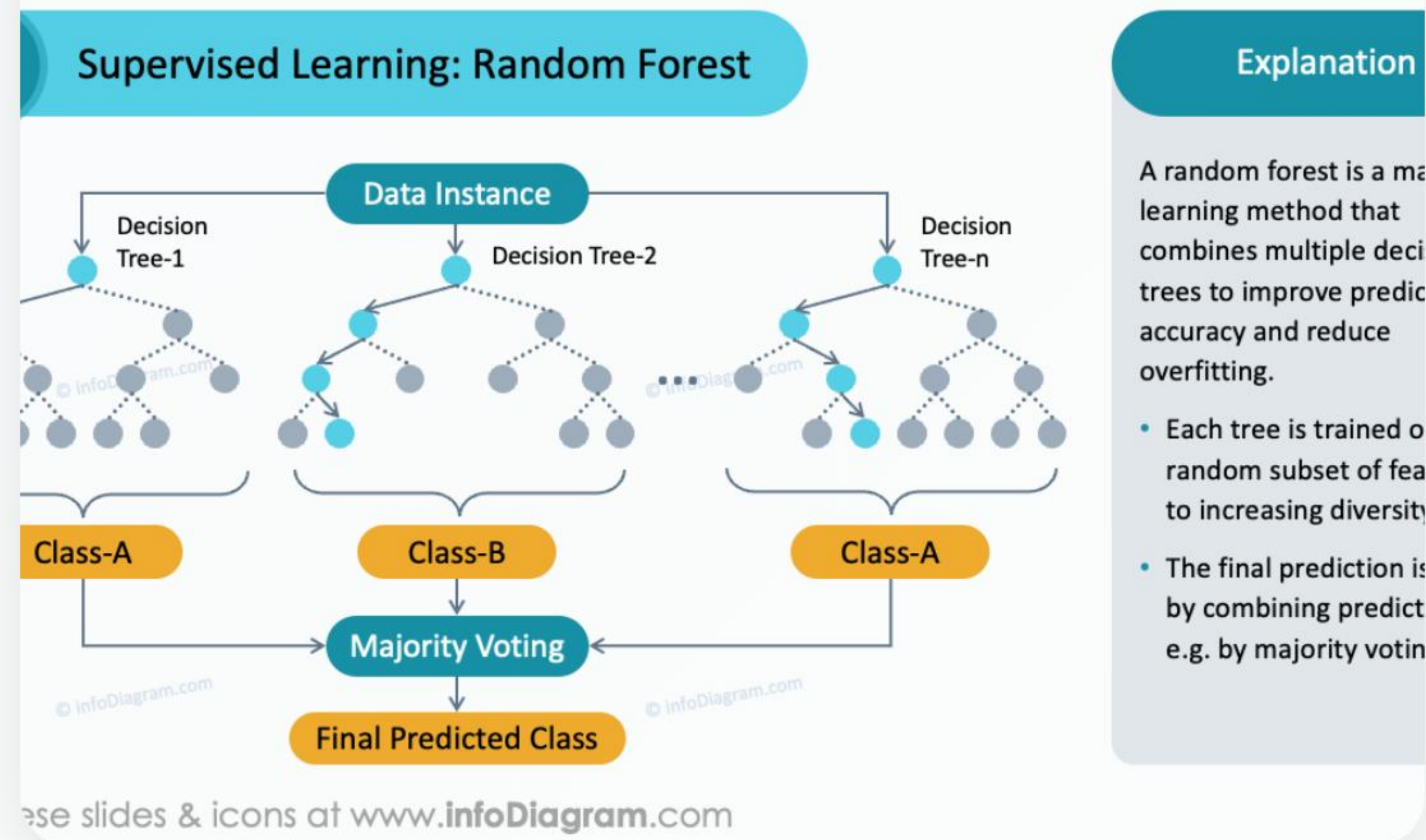
Class 1 (High Perf)

The dataset exhibits a healthy balance between categories (Score < 70 vs Score $\geq$ 70), ensuring model neutrality.

Why Random Forest?

Random Forest Illustration - Supervised Learning Algorithm

Machine Learning AI Model Flowchart Diagram, Decision Tree Set, Predicted Class, Voting



Ensemble Power

Random Forest builds multiple decision trees and merges them together to get a more accurate and stable prediction.

Bagging Strategy: It reduces variance without increasing bias, making it perfect for educational datasets with diverse categorical outliers.

Accuracy Benchmark

98%

Testing Accuracy Achieved

The model demonstrates near-perfect precision across the test set (200 records), successfully identifying high-performers with minimal false positives.

Classification Matrix Breakdown



True Negatives

100 Cases correctly identified as Class 0 (Below Threshold).



True Positives

96 Cases correctly identified as Class 1 (Above Threshold).



Total Errors

Only 4 misclassifications across the entire test partition.

Questions?

Predictive Modeling for Student Success

Project Reference: [StudentsPerformance.csv](#)

Technology: Python | Random Forest | scikit-learn

Image Sources



Thumbnail for

Source: [www.vecteezy.com](https://static.vecteezy.com/system/resources/previews/072/324/607/non_2x/diverse-group-of-university-students-studying-and-learning-in-a-traditional-library-setting-photo.jpg)

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